

Anvesh Reddy

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EDUCATION

University of Illinois Urbana-Champaign

Grainger College of Engineering | B.S. in Engineering Mechanics (Concentration in Robotics),
Minor in Computer Science, Minor in Mathematics

Expected Graduation: May 2028

GPA: 3.6/4.0

TECHNICAL SKILLS

Coursework: Statics, Dynamics, Solid Mechanics, Thermodynamics, Fluid Mechanics, Computational Mechanics, Design for Manufacturability, Engineering for Global Sustainability, Introduction to Robotics, Humanoid Robotics

CAD & Design: Fusion 360, SolidWorks, Onshape, ANSYS, aPriori, 3D Printing, FEA, Parametric design optimization,

Software: Python, C++, Git/GitHub, Linux, VS Code, Docker, ROS 2, Gazebo, Nav2, SLAM Toolbox, TurtleBot 3, ML

WORK EXPERIENCE

Derek Hoiem Computer Vision Lab | Undergraduate Researcher

August 2026 - Present

Champaign, Illinois

- Evaluating 4 VLMs on 15 ALOHA rollouts to identify and label reusable sub-trajectories from partial and failed trials
- Building an API pipeline to ingest, segment, and label ALOHA videos for self-supervised VLA fine-tuning

SigRobotics | Mechanical Design Engineer

May 2026 - Present

Champaign, Illinois

- Designing four articulated fingers for the K1 humanoid with dual-motor flexion-extension and single-motor abduction.
- Optimizing components for 3D printing while coordinating assembly compliance with electrical, palm & forearm teams

IT Service Desk | Team Lead

October 2025 – August 2026

Champaign, Illinois

- Resolved 750+ hardware and software support tickets before being promoted to IT Support Team Lead
- Led staffing and ticket assignments for 50 consultants across three locations, maintaining 97.7% customer satisfaction

TECHNICAL PROJECTS

Mechanical Gripper | Mechanical Design

- Designed and fabricated a 3D-printed, nine-part parallel-jaw gripper producing 17.7 N grip force from a 6 V servo through a three-gear, 1.5:1 reduction
- Ran a six-point parametric FEA study of jaw thickness and geometry, reducing mass by 47.5% while maintaining a 4.86× factor of safety; built a regression model to predict stress and displacement under load
- Produced 5 production-ready engineering drawings with GD&T, datums, and tolerances per ASME Y14.5-2018

Handheld Electromechanical Device | Mechanical Design

- Designed a sub-85 g ABS enclosure with snap-fit assembly and internal ribbing for a DC motor and drivetrain.
- Iterated over 4 prototypes and used digital calipers, feeler gauges and tolerance-stack analysis to identify and correct a 0.45 mm motor-slot deviation
- Developed 3D-printing and injection-molding variants; aPriori DFM analysis reduced projected molding cost by 17%.

UR3 Robotic Arm Drawing Pipeline | Robotics & Autonomy

- Developed a closed-form 6-DOF inverse-kinematics solver for a UR3 robotic arm using Denavit–Hartenberg parameters, converting target end-effector poses into joint commands for physical-hardware execution
- Built a ROS/Python perception-to-actuation pipeline combining blue-channel segmentation, Canny edge detection, contour extraction, coordinate transformation and inverse kinematics to autonomously draw three test images
- Calibrated an image-to-robot coordinate transformation across a 0.35 m × 0.28 m drawing workspace and implemented nearest-neighbor contour sequencing to reduce unnecessary end-effector travel by 17% while drawing images

Autonomous Robot Exploration | Robotics & Autonomy

- Developed an autonomous exploration pipeline for TurtleBot3 in ROS 2/Gazebo using slam_toolbox for real-time SLAM, Nav2 for navigation, and a custom frontier detection algorithm with BFS-based clustering
- Achieved a 100% completion rate across 10 trials with randomized starting poses, averaging 2.9 minutes per run and 5.1% map coverage per meter traveled
- Automated unattended launch, teardown, and performance logging for the full ROS 2 stack; prevented navigation loops using progress-based frontier blacklisting and Nav2 recovery-triggered goal cancellation